# FEB12001X - Exam 27 October 2023

FEB12001X Applied Microeconomics 23/24 (ESE) · 9 exercises · 52.0 points

# 1 Exercise 1

8.0 points · 3 questions

The government of Highland aims for providing good public service while also considering the costs. The government's objective function is as follows: $N\left(W\right)\cdot Q-\mu\cdot W\cdot N\left(W\right)$ where $N\left(W\right)$ is the number of employees in public service, $Q$ is the quantity of services each employee produces, $W$ is the salary of a public service employee and $\mu$ is a parameter which indicates the government's emphasis on costs. The number of employees in public service increases with the salary $W$, in other words the first derivative of the function $N\left(W\right)$ is positive. The government maximizes its objective function. You may assume that the second-order condition is satisfied.

Text

a Derive the first-order condition which describes the optimal choice of the salary $W$ for the government. For each of the three terms of the first-order condition, provide the economic meaning in one clear sentence (no "math in words", but an explanation which a non-economist could also understand).

4.0 points · Open · 1/2 Page

**+1 point**

The first order condition is: $N'(W)Q - \mu N(W) - \mu W N'(W) = 0$
Instead of $N'(W)$, $\frac{dN}{dW}$ is also correct.

**+1 point**

The first term shows the effect of a higher salary on the number of employees and hence on the total production.

**+1 point**

The second term shows that a higher salary is costly because the employees who are already working are being paid a higher salary.

**+1 point**

The third term shows that a higher salary is costly because the extra employees who will join the organisation will be paid a salary

b Suppose that due to a technological innovation the productivity of employees in public services increases (that is: $Q$ increases). In response to this, will the government increase or decrease the number of public service employees? Use the first-order condition. But also explain why the government will do this, without using mathematical terms.

2.0 points · Open · 1/2 Page

**+1 point**

From the first order condition we can see that an increase in $Q$ means that the benefits of a wage increase are rising (see first term). Hence, the government will set a higher salary and employ more employees.

**+1 point**

The reason is that employing more employees will lead to a bigger increase of valuable production if the productivity is higher.

c The magnitude of the first derivative of the function $N\left(W\right)$ depends on the degree of monopsony power of the employer. Explain, in a few clear sentences, why this is the case.

2.0 points · Open · 1/2 Page

**+1 point**

Monopsony power means that the employer has some freedom in setting the salary. If he lowers the salary, not all employees will leave directly. If the salary rises, the increase in the number of employees is also limited.

**+1 point**

The higher the $N'\left(W\right)$, the stronger the response of the labour supply to the salary at this employer, and the smaller the monopsony power. And vice versa of course.

## 2 Exercise 2

6.0 points · 3 questions

A company is looking for new employees. There are two types of employees in the labour market: some employees are intrinsically motivated, others are not. The utility function of an employee who is not intrinsically motivated is as follows: $U = Y - \theta e^2$, where $U$ is the utility of the employee, $Y$ the salary, and $e$ the efforts made by the employee on the job. If an employee chooses another job, his utility is equal to $\overline{U}$. This applies to both an intrisically and a non-intrinsically motivated employee.

Text

a What does the utility function of an intrinsically motivated employee look like? Explain your answer with a clear sentence.

2.0 points · Open · 1/4 Page

### +1 point

For example: $U = Y - \theta e^2 + \gamma e$ where $\gamma > 0$. But every extra term which is rising in efforts $e$ is fine.

### +1 point

The additional term ensures that the employee is willing to make efforts without an external motivator like a bonus or recognition.

b Suppose the company offers a salary which is equal to the fixed amount $X$. Under what conditions would a non-intrinsically motivated employee want to work at this company? In your answer, make use of the employee's choice of effort.

2.0 points · Open · 7/20 Page

**+1 point**

The employee has to get at least as much utility at this company as any other, thus $X - \theta e^2$ must be larger or equal to $\overline{U}$.

**+1 point**

A non-motivated employee will make zero effort, so the condition will become $X$ larger or equal to $\overline{U}$.

c Suppose the company wants to employ intrinsically motivated employees, but the intrinsic motivation of an applicant is not visible for this company. Can the company, by choosing the salary level $X$ in a smart way, achieve self-selection of intrinsically motivated employees? In your answer, make use of the utility function you proposed in part a of this exercise.

2.0 points · Open · 1/2 Page

**+1 point**

Yes, a basesalary that is not too high (but also not too low) can achieve self-selection of intrinsically motivated employees.

**+1 point**

More precisely, a base $X$ salary for which the following conditions hold:
$X$ is smaller than $\overline{U}$ and $X$ is larger or equal to $\overline{U} - \frac{\gamma^2}{4\theta}$. This follows from deriving the optimal efforts for both types of employees, filling this into the utility functions and writing down the participation constraints.

# 3 Exercise 3

10.0 points · 3 questions

Two freelancers are thinking about working together. If they work alone, they each achieve a revenue equal to $p\kappa e_i$, where $p$ is the price of the product, $e_i$ the efforts that freelancer $i$ makes and $\kappa$ is a parameter which represents the freelancer's productivity. If they work together, they achieve a combined revenue equal to $p\kappa\beta\left(e_1+e_2\right)$, where $e_1$ are the efforts of freelancer 1, $e_2$ the efforts of freelancer 2, and $\beta>1$ is a technical parameter. The utility function of freelancer $i$ is equal to the sum of his income minus the cost of effort $\theta e_i^2$. If the freelancers work alone, the income of a freelancer is equal to his revenue. If they work together, they share the revenue (each has an income equal to half of the revenue). In all situations (working alone or working together):

1. Each freelancer maximizes his own utility.
2. Each chooses his own efforts independent of the other.
3. They cannot make any agreements about their efforts beforehand.

Text

a Derive under what condition working together will lead to a higher income for both freelancers? Also write two sentences, explaining the two reasons why the income changes due to working together.

4.0 points · Open · 13/20 Page

## +2 points

If they work alone, each maximizes utility $p\kappa e_i - \theta e_i^2$. The first order condition is $p\kappa - 2\theta e_i = 0$ and thus $e_i = \frac{p\kappa}{2\theta}$. The income is $p\kappa\left(\frac{p\kappa}{2\theta}\right)$, which is $\frac{(p\kappa)^2}{2\theta}$. If they work together, each maximizes utility $\frac{1}{2}\left(p\kappa\beta e_i + p\kappa\beta e_j\right) - \theta e^2$. The first order condition is $\frac{1}{2}p\kappa\beta - 2\theta e = 0$ and thus $e = \frac{p\kappa\beta}{4\theta}$. The income will then be $\frac{1}{2}p\kappa\beta\left(\frac{p\kappa\beta}{4\theta} + \frac{p\kappa\beta}{4\theta}\right) = \frac{(p\kappa\beta)^2}{4\theta}$.

## +1 point

Comparing the income in both situations will lead to (after some rewriting): if $\beta > \sqrt[2]{2}$ then collaborating leads to a higher income.

## +1 point

On the one hand, working together will lead to more income due to synergy effects $(\beta > 1)$ (the word synergy is not necessary, an explanation of it is also fine). On the other hand, there is an externality problem (free-rider problem): the freelancers are inclined to make less efforts, because the benefits from it will be shared.

b Derive under what condition the freelancers will decide to work together.

4.0 points · Open · 1/2 Page

**+1 point**

The utility that each freelancer achieves by working alone is equal to $p\kappa e - \theta e^2$, where $e = \frac{p\kappa}{2\theta}$ and thus $\frac{(p\kappa)^2}{4\theta}$.

**+1 point**

The utility that each freelancer achieves by working together is $\frac{1}{2}p\kappa\beta\left(\frac{p\kappa\beta}{4\theta} + \frac{p\kappa\beta}{4\theta}\right) - \theta\left(\frac{p\kappa\beta}{4\theta}\right)^2$, which is $\frac{3}{16}\frac{(p\kappa\beta)^2}{\theta}$.

**+2 points**

Comparing the utilities shows that working together will lead to higher utility if $\beta > \sqrt[2]{\frac{4}{3}}$.

c Now suppose the freelancers are friends and care about each other's utility (i.e. they are altruistic). Formulate a utility function for freelancer 1 which reflects this altruism towards freelancer 2. In your answer, build on the utility function as described above.

2.0 points · Open · 1/4 Page

**+2 points**

A possible answer is: $U_1 = p\kappa e_1 - \theta e_1^2 + \rho U_2$. But a utility function where agent 1 cares about the income and the cost of effort of agent 2 with a weight is also great. Likewise, a utility function where agent 1 cares about the income of agent 2 with a weight is correct.

## 4 Exercise 4

2.0 points · 1 question

A HR-manager of a big company wants to know if there is discrimination in the recruitment of personnel. The company has a thousand branch offices that frequently recruit employees through advertising vacancies and having interviews with applicants.

Text

How can the HR-manager most reliably answer the question of whether there is discrimination in recruitment within his company? Describe a research design in no more than 4 sentences.

2.0 points · Open · 2/5 Page

### +2 points

By conducting a correspondence test, in which a large number of fake but realistic-looking job applications are sent out, and then examining whether the responses from recruiters depend on characteristics of the applicant that should not matter.

# 5 Exercise 5

7.0 points · 3 questions

Yasmine earns an hourly wage of 20 euro in a perfectly competitive market. She has 200 hours available per month to divide between work and leisure. She maximises her utility, which is given by

$$U = Y + 400ln(L)$$

where $Y$ is Yasmine's net income in euro and $L$ is the number of hours of leisure Yasmine chooses to consume per month. The government implements a proportional income tax with a marginal tax rate of 50 percent.

Text

a Derive how many hours of leisure Yasmine chooses to consume per month.

3.0 points · Open · 7/10 Page

## +1 point

Budget constraint is Y=(1-0.5)∗20∗(200-L)= 2000-10L.

## +2 points

Substituting in utility function and maximising wrt L yields first-order condition -10+((400)/L)=0→L=((400)/(10))=40. Using Lagrange is possible as well.

b For Yasmine, the equivalent variation of the income tax is 1722,74 euro.
Determine the excess burden of the income tax.

2.0 points · Open · 1/2 Page

### +1 point

Under the income tax, Yasmine works 160 hours, earns gross 3200 euro and pays 1600 euro in taxes.

### +1 point

Given that the EV = 1722.74, the excess burden is 1722.74-1600=122.74 euro.

### 0 points

Beyond the question: You can calculate the equivalent variation as follows: Without tax, the budget constraint is $Y=20*(200-L)= 4000-20L$. Substituting in utility function and maximising to L yields first-order condition $-20+((400)/L)=0 \rightarrow L=((400)/(20))=20$. Hence, she works 180 hours and earns gross 3600 euro. We can determine the equivalent variation by equating utility from the income tax with the utility with lump-sum tax T (=equivalent variation):
$1600+400\ln(40)=3600-T+400\ln(20)$, which can be solved for T=1722.74

Now consider Bob, who lives in the same country as Yasmine. Bob also chooses between working and leisure. Bob has diminishing marginal utility of both income and leisure. Under the income tax, Bob chooses to work 100 hours per month. In the absence of an income tax, Bob would also choose to work 100 hours per month.

Text

c Does the income tax levied on Bob create an excess burden? Explain your answer in maximally four sentences.

2.0 points · Open · 3/10 Page

### +1 point

Correct link between substitution effect and excess burden, but incorrect conclusion that there is no substitution effect here.

### +2 points

Yes. The income tax changes the relative prices of income and leisure, implying there is a substitution effect (towards leasure). This leads to an Excess Burden.

### 0 points

Beyond the question: Given diminishing marginal utility, there is also an income effect of the income tax. Apparently, for Bob the income and the substitution effect on the choice of leisure cancel out.

# 6 Exercise 6

7.0 points · 3 questions

In deriving the Ramsey rule (or Inverse elasticity rule), we assumed that the government could not implement a lumpsum tax.

Text

a Explain why we had to make that assumption in order to derive the Ramsey rule.

2.0 points · Open · 3/10 Page

**+1 point**

A lumpsum tax has no excess burden.

**+1 point**

The Ramsey rule is derived assuming that the government aims to minimise the loss in efficiency due to taxation. Hence, if a lumpsum tax is available, the government would only use the lumpsum tax.

Consider a government that needs to collect tax revenue to finance an infrastructural project. It introduces ad valorem taxes $t_X$ and $t_Y$ on good $X$ and good $Y$, respectively. Supply of both good $X$ and good $Y$ is given by $P = MC = 100$. Demand for good $X$ is given by

$$P_X = 400 - \frac{1}{2}Q_X$$

while demand for good $Y$ is given by

$$P_Y = 300 - \frac{1}{2}Q_Y$$

The total tax revenue collected is exactly equal to the construction cost of the infrastructural project. The government uses the Ramsey rule to determine the tax rates and optimally sets $t_X = 0.3$.

Text

b Derive the optimal level of $t_Y$.

2.0 points · Open · 3/5 Page

**+1 point**

Demand elasticity $\varepsilon_X = 2((100)/(600)) = 1/3$. Demand elasticty $\varepsilon_Y = 2((100)/(400)) = 1/2$ .

**+1 point**

Using the inverse elasticty rule we get $t_X/t_Y = \varepsilon_Y/\varepsilon_X \rightarrow t_Y = 0.3((1/3)/(1/2)) = 0.2$.

c Derive how much value the infrastructural project should at least generate in order for it to be an efficient investment by the government, given that the project is financed as described above. Note: if you were not able to derive $t_Y$ in question b), assume a particular level of $t_Y$ here.

3.0 points · Open · 3/5 Page

**+1 point**

Tax revenues: Before tax equilibrium market X has Q=600. Tax is 30 euro per unit. After-tax Q=540. Hence, tax revenue equals 30*540 = 16200. Before tax equilibrium market Y has Q=400. Tax is 20 euro per unit. After-tax Q=360. Hence, tax revenue equals 20*360 = 7200.

**+1 point**

Excess burdens: Before tax equilibrium market X has Q=600. Tax is 30 euro per unit. After-tax Q=540. Hence, EB = (1/2)30$*$60= 900. Before tax equilibrium market Y has Q=400. Tax is 20 euro per unit. After-tax Q=360. Hence, EB = (1/2)20$*$40= 400.

**+1 point**

The benefits must be larger than the economic cost, which is total tax revenue plus total excess burden. Hence, total economic cost are 16200 + 7200 + 900 + 400 =24700 euro.
Alternative method: calculate the reduction in consumer surplus in both markets.

# 7 Exercise 7

10.0 points · 4 questions

Consider a village located at the sea with N citizens. Each citizen maximises own earnings. Citizens must choose between working in a factory and working as a fisherman. Working in a factory yields earnings of $w$ euro. The earnings of a fisherman depend on the amount of fish caught which, in turn, depends on the total number of citizens that chooses to become a fisherman. The earnings $E$ of each individual fisherman are given by

$$E = a - bF$$

where $F$ is the total number of citizens that chooses to become a fisherman. Each citizen decides independently whether to work in a factory or as a fisherman. Assume that $b > 0$ and that in any equilibrium some citizens work in a factory and some citizens become fisherman.

Text

a Derive the number of fishermen in equilibrium.

2.0 points · Open · 3/10 Page

+2 points

Citizens choose to fish if E≥w, so that in equilibrium a-bF-w=0→F=(a-w)/b.

Suppose the government maximises the sum of citizens' earnings. It regulates the fishing industry by requiring each fisherman to buy a permit. Without a permit, a citizen is not allowed to work as fisherman.

Text

b Derive the socially efficient number of fishermen. Explain in two sentences why this differs from the answer to question a).

4.0 points · Open · 7/10 Page

**+1 point**

Maximising the sum of earnings FE+(N-F)w→F(a-bF)+(N-F)w with respect to F yields the socially efficient number of fishermen

**+1 point**

This gives first-order condition a-bF-w-bF=0→F=(a-w)/2b

**+2 points**

The difference with a) arises because each individual fisherman does not take into account the negative externality imposed on other fishermen. This lead to too many fishermen in equilibrium, a problem known as the Tragedy of the Commons.

c Show that the optimal price of a permit is equal to $\frac{1}{2}(a-w)$.

2.0 points · Open · 3/5 Page

**+1 point**

We are effectively looking for a Pigouvian tax, which is equal to the negative external cost at the socially efficient level of F. External cost is equal to the difference between private cost and benefits given by a-bF-w and social cost and benefits given by a-bF-w-bF as derived under a) and b) respectively. Hence, external cost are bF.

**+1 point**

Socially efficient level of F=((a-w)/(2b)). Substituting gives the optimal price of a permit: ((a-w)/2).

**+2 points**

Alternative solution method: find permit price p such that a-bF-p=w yields F=((a-w)/2b. Substituting gives the optimal price of a permit: ((a-w)/2).

d The previous question shows that the optimal price of a permit decreases in $w$. Explain why in three sentences.

2.0 points · Open · 3/10 Page

**+1 point**

If w increases, fewer people choose to become a fisherman.

**+1 point**

As the marginal external cost decrease if the number of fishermen decreases, a lower permit price is necessary to correct the market failure.

# 8 Exercise 8

2.0 points · 1 question

Consider a farmer who owns 100 cows. The cows impose a negative externality on the local community. The local economist has determined that the socially efficient number of cows equals 80. The economist advises the local government to intervene in one of the following two ways:
1. Impose a Pigouvian tax per cow on the farm.
2. Offer a Pigouvian subsidy per cow removed from the farm.
The government decides to implement the Pigouvian subsidy.

Text

Give one ***normative*** reason and one ***positive*** reason for implementing the Pigouvian subsidy rather than the Pigouvian tax. Clearly indicate which of your explanations is normative and which is positive.

2.0 points · Open · 2/5 Page

### +1 point

Normative: Distributional arguments: the farmer is relatively poor or has relatively high weight in the social welfare function. Note: efficiency is not a valid argument.

### +1 point

Positive: Any answer that involves (influencing of) politicians and/or voters is valid, including lobbying, rent-seeking, bribing, intimidating, and a majority of villagers who like the farmer. Normative answers (linking to efficiency or disttribution) are wrong.

## 9 Additional answer space

0.0 points · 5 questions

Only use this extra space if you made a mistake in one of your answers. Please do not forget to **indicate in the answer box of that particular question where the correct answer can be found**. Otherwise, the answers in the additional space below will not be reviewed.
NB: Do **not** use this additional space for answers that are too long.

Text

a Additional answer box 1

Open · 4/5 Page

b Additional answer box 2

Open · 1/2 Page

c Additional answer box 3

Open · 1/2 Page

d Additional answer box 4

Open · 1/2 Page

e Additional answer box 5

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